

REFLECTION AS A CATEGORICAL BRIDGE: CERTIFYING SOURCE-TO-SOURCE
TRANSFORMATIONS BETWEEN PROCEDURAL AND DATAFLOW SEMANTICS

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ABSTRACT

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CHAPTER 1

Introduction

1.1. Context

1.2. The Problem

1.3. Why this is hard

1.4. Two underused tools

1.5. Contributions

CHAPTER 2

Theoretical Foundations

- 2.1. Operational and denotational semantics for a small imperative core
- 2.2. Dataflow / graph IR semantics
- 2.3. Computational reflection
- 2.4. Categorical preliminaries
- 2.5. Semantic preservation

CHAPTER 3

Literature Review

TODO (?)

CHAPTER 4

Analysis

TODO

CHAPTER 5

New Proposal

TODO

CHAPTER 6

Conclusion

TODO

BIBLIOGRAPHY